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**Classification of Eye Diseases from OCT
and OCTA Images
Using Deep Learning Models**

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Preface

This graduation project was prepared within the scope of the ISE 402 Graduation Project course in the Department of Software Engineering. We would like to thank our advisor, Prof. Dr. Ünal Çavuşoğlu, for guiding us with his knowledge and experience throughout the study and for his valuable contributions.

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Abstract

Early and accurate identification of retinal diseases is important for preventing vision loss and planning appropriate treatment. This study investigates the classification of normal cases, diabetic retinopathy (DR), and age-related macular degeneration (AMD) from optical coherence tomography (OCT) and optical coherence tomography angiography (OCTA) images using deep learning methods. Numerous experiments involving different training and decision settings were conducted during development. For each architecture–imaging modality combination, a selected model configuration was reported by comparing the test classification reports of ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 on OCT and OCTA data. The results of eight selected model configurations are therefore presented comparatively.

A common patient-level data split was used to prevent images belonging to the same patient from being distributed across the training, validation, and test sets. Weighted sampling, class weights, Focal Loss, and data augmentation methods were used to reduce the effect of class imbalance. Model performance was evaluated at both image and patient levels using accuracy, balanced accuracy, precision, recall, Macro-F1, and confusion matrices. Patient-level decisions were obtained by aggregating the class probabilities produced for images belonging to the same patient.

In the descriptive results of the selected configurations, the OCT ResNet18-based run achieved 96.53% accuracy, 96.19% balanced accuracy, and 95.14% Macro-F1 at the image level. The OCTA DenseNet121-based run achieved 94.70% accuracy, 90.62% balanced accuracy, and 89.58% Macro-F1 at the image level, and 97.44% accuracy and 96.46% Macro-F1 at the patient level. These values belong to selected configurations with different preprocessing, augmentation, loss, and decision settings; they do not demonstrate that the ResNet18 or DenseNet121 architecture is structurally superior to the other architectures.

The findings indicate that deep learning models have potential for computer-aided classification of eye diseases from OCT and OCTA images. The reported results belong to configurations selected by examining test classification reports, while decision thresholds in threshold-based configurations were determined on the validation set. Because the test set was used for configuration selection, these values should be interpreted as descriptive performance of the selected configurations on the current dataset rather than as an independent and unbiased final test estimate. Furthermore, the presence of only one AMD patient in the test sets limits the generalizability of the patient-level AMD results in particular. The findings therefore do not constitute a direct clinical diagnostic claim; instead, they present an experimental decision-support approach that should be validated on larger, independent, and class-balanced datasets.

Keywords: Deep learning, OCT, OCTA, diabetic retinopathy, age-related macular degeneration, image classification.

1 Introduction

The retina is one of the ocular tissues that plays a fundamental role in perceiving visual information and converting it into neural signals. Diabetic retinopathy (DR) and age-related macular degeneration (AMD) can cause structural or vascular changes in the retina and may lead to severe vision loss at advanced stages. Optical coherence tomography (OCT), which visualizes the cross-sectional structure of retinal tissue, and optical coherence tomography angiography (OCTA), which provides non-invasive visualization of the retinal vasculature, are important sources of information for evaluating these diseases [1, 2].

Because OCT and OCTA examinations produce many images, expert evaluation can be time-consuming. The variation of disease findings according to imaging modality, disease stage, and image quality further complicates this process. Computer-aided image analysis methods can support the consistent examination of image patterns and the identification of possible disease classes without replacing expert judgment.

Deep learning is a machine learning approach that learns hierarchical features from data through multilayer artificial neural networks [3]. Convolutional neural networks are widely used in medical image analysis because they can automatically learn local and spatial patterns in images. The ResNet, DenseNet, and EfficientNet families are prominent architectures that offer different connectivity and scaling strategies for image classification problems [4, 5, 6].

1.1 Problem Definition

The main problem addressed in this study is the classification of OCT and OCTA retinal images into the *NORMAL*, *DR*, and *AMD* classes. Patient counts are markedly imbalanced across the classes. In particular, the small number of patients in the *AMD* class means that evaluations based only on overall accuracy may not adequately reflect performance on minority classes.

The presence of many images per patient creates a risk that images from the same patient may be distributed across the training, validation, and test sets if splitting is performed randomly at the image level. Such data leakage can make the model’s performance on new patients appear higher than it actually is. The data were therefore split by patient identifiers, and the results were examined at both image and patient levels.

The objective is not merely to obtain high overall accuracy, but to examine the classification behavior of selected model configurations while considering class imbalance, patient dependence, and different imaging modalities. Because the preprocessing, augmentation, loss functions, and decision methods of the experimental configurations are not identical, the results are not treated as a controlled comparison of architectural superiority.

1.2 Aim of the Study

The main aim of the study is to descriptively examine image- and patient-level classification results of selected experimental configurations based on ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 for distinguishing the *NORMAL*, *DR*, and *AMD* classes in OCT and OCTA images. The following objectives were defined:

- develop and train model configurations based on ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 separately on OCT and OCTA data,
- use a common patient-identifier split to prevent images belonging to the same patient from being distributed across different datasets,
- apply methods such as weighted sampling, class weights, Focal Loss, and MixUp in selected configurations to manage class imbalance,
- evaluate results using balanced accuracy, Macro-F1, and class-level precision, recall, and F1 scores in addition to accuracy,
- generate patient-level predictions by aggregating image probabilities from the same patient with configuration-specific methods,
- report the results of eight selected experimental configurations separately for OCT and OCTA,
- explicitly state that, because of configuration differences, the results do not constitute a controlled comparison of architectural or imaging-modality superiority.

1.3 Research Questions

The study addresses the following research questions:

1. What image- and patient-level results were observed in the eight selected experimental configurations?
2. How do configuration-specific preprocessing, augmentation, and decision methods limit the interpretation of the results?
3. How do accuracy, balanced accuracy, and Macro-F1 differ under class imbalance?
4. How does patient-level aggregation of image probabilities affect classification results?

1.4 Contributions and Scope

The main contribution of the study is the detailed documentation of eight selected experimental configurations based on four convolutional neural network architectures across two retinal imaging modalities. The use of a common patient-identifier split enabled experiments within the same modality to be conducted on the same available patient groups. Because two patients had no OCT folder, patient counts in the OCT and OCTA sets are not completely identical. In addition to architecture, the configurations differ in input size, augmentation, loss function, decision threshold, and patient-level aggregation. The study is therefore not a controlled ablation intended to determine which of ResNet18, DenseNet121, or EfficientNet is structurally superior. The reported values represent the observed outcome of the complete training and evaluation pipeline together with the architecture.

The study does not claim to develop a clinical diagnostic system. The models were evaluated for computer-aided classification on a limited and imbalanced dataset. In particular, the presence of only one AMD patient in the test set is a major limitation when interpreting patient-level AMD performance and the overall results.

1.5 Organization of the Thesis

The following chapter first presents the fundamental concepts and literature related to OCT, OCTA, the retinal diseases considered, and the deep learning approaches used. Data preparation, patient-level splitting, augmentation, model training, and evaluation methods are then explained. The proposed experimental system chapter describes the software workflow and configuration-dependent components, while the experimental results chapter examines the eight selected experiments individually. The final chapter evaluates the observed findings, contributions, and limitations of the study.

2 Fundamental Concepts and Literature Review

2.1 Optical Coherence Tomography

Optical coherence tomography is a non-invasive imaging method that uses low-coherence interferometry to produce high-resolution cross-sectional images of biological tissues. The fundamental principles of OCT were introduced by Huang et al., who demonstrated that the method enables micrometer-scale imaging of different tissues, including the retina [1]. In retinal imaging, OCT is used to examine retinal layers, fluid accumulation, tissue thickness, and various structural abnormalities.

OCT volumes may consist of many consecutive cross-sectional images from the same patient. Although this provides a rich data source for deep learning models, the high similarity between images increases the importance of patient-level data splitting in experimental design. Distributing slices from the same volume or patient across training and test sets may produce results that do not reflect the model’s true generalization performance.

2.2 Optical Coherence Tomography Angiography

OCTA uses motion and decorrelation information from consecutive OCT scans to visualize retinal and choroidal vasculature without contrast agents. Split-spectrum amplitude-decorrelation angiography is one of the fundamental methods used to extract flow signals from OCT data [2]. OCTA images can support the evaluation of vascular characteristics such as vessel density, vessel continuity, and avascular regions.

While OCT emphasizes structural tissue information, OCTA focuses on vascular patterns. The same disease classes may therefore be represented by different visual cues in the two modalities. Evaluating models separately on OCT and OCTA data enables examination of how the reported classification performance varies by imaging modality.

2.3 Diabetic Retinopathy and Age-Related Macular Degeneration

Diabetic retinopathy is associated with damage caused by diabetes to retinal blood vessels. Findings may include microvascular changes, hemorrhage, edema, and abnormal vessel formation at advanced stages [7]. Age-related macular degeneration is a progressive retinal disease that affects the macula, which is responsible for central vision. Drusen deposits, atrophy, and neovascular changes may be observed in different forms of the disease [8, 9].

Automatic classification is challenging because the appearance of these diseases may vary according to the patient, disease stage, and imaging method. In datasets where normal samples outnumber diseased samples, the model may also become biased toward the majority class. Class-level recall and metrics such as Macro-F1 should therefore be evaluated together with overall accuracy.

2.4 Convolutional Neural Networks and Transfer Learning

Convolutional neural networks consist of filters that learn local image features such as edges, textures, and shapes. Deeper network layers obtain more complex and class-specific representations [3]. Because labeling medical image datasets requires expertise, transfer learning, which adapts models pretrained on large natural-image datasets to medical images, is widely used.

ResNet uses residual connections to reduce optimization problems in deep networks [4]. DenseNet provides dense connections from each layer to subsequent layers to encourage feature reuse [5]. EfficientNet jointly scales network depth, width, and image resolution to produce models that balance accuracy and computational cost [6]. The different design characteristics of these architectures may produce different results when learning structural and vascular patterns in OCT and OCTA images.

2.5 Deep Learning Studies on Retinal Images

Studies on retinal images have shown that deep learning can be used for eye-disease classification and referral. Gulshan et al. evaluated a deep learning algorithm for detecting referable diabetic retinopathy from fundus photographs on two clinical datasets [7]. Although the study focused on fundus images, it is important because it demonstrated the potential of deep convolutional networks for automated analysis of retinal diseases.

Kermany et al. used a transfer-learning approach to distinguish normal, choroidal neovascularization, diabetic macular edema, and drusen classes in OCT images [8]. Their study showed that pretrained deep networks can be effective for multiclass OCT classification. De Fauw et al. developed a two-stage system that combined tissue segmentation with clinical referral decisions in three-dimensional OCT scans and evaluated it on a large clinical dataset [9].

2.6 Disease Classification Studies on OCTA-500

The images used in this study belong to the OCTA-3mm subset of the OCTA-500 dataset introduced by Li et al. [10]. OCTA-500 contains paired OCT and OCTA volumes from 500 patients, projection images generated from different retinal layers, demographic information, disease labels, and various segmentation labels. The subset with a 3 mm field of view consists of 200 patients and includes the NORMAL, DR, AMD, and CNV classes. The primary experimental focus of the original dataset study was joint segmentation of capillaries, arteries, veins, and the foveal avascular zone rather than disease classification.

Lin et al. addressed FAZ segmentation and disease classification within a joint learning framework called BSDA-Net [11]. They used the NORMAL, DR, and AMD classes of OCTA-500 and combined resized en-face projection images obtained from 3 mm and 6 mm fields of view. Features derived from the FAZ boundary, shape, and distance map were transferred to the classifier, and five-fold cross-validation was applied. BSDA-Net achieved 94.23% accuracy and 91.69% Macro-F1 on OCTA-500. This result indicates that

FAZ-related features may support disease classification. However, because the data unit was an en-face projection image, the result is not directly comparable with the image-level results in this thesis, where volume slices are used as separate prediction units.

Aydin et al. compared ResNet50-based OCTA classification with ResNet50- and RETFound-based OCT classification for the NORMAL, DR, and AMD classes on OCTA-500 [12]. Stratified cross-validation was used to manage class imbalance, and a four-class experiment grouped different diseases under an “other” category. In the three-class experiment, ResNet50-OCTA outperformed ResNet50-OCT but achieved lower classification performance than RETFound-OCT, which had been pretrained on a large OCT dataset. This study is close to the present thesis in comparing OCT and OCTA inputs from the same data source; however, the projections, backbones, and validation protocol differ.

Zhang et al. proposed the Multi-step Interaction Network (MINet), which jointly evaluates OCT and OCTA volumes [13]. The model converts three-dimensional OCT and OCTA data into learnable two-dimensional representations and interactively combines shared and complementary features between the two modalities. In a seven-class experiment on OCTA-500, it achieved 86.90% accuracy and a 70.81% F1 score. MINet’s integration of both modalities in a single model is its main methodological difference from this study, which evaluates OCT and OCTA in separate experimental pipelines.

Shah et al. investigated cross-modal transfer learning from retinal foundation models for OCTA disease classification under limited-sample conditions [14]. Four classes were used on OCTA-500: NORMAL, DR, AMD, and other diseases. Superficial and deep OCTA projections were combined using intermediate feature fusion. Fifteen percent of the data was reserved as an independent test set, and five-fold cross-validation was applied to the remainder. Accuracy of 81.33% was reported for the VisionFM-based model and 80.00% for the RETFound-based model, with domain-specific pretraining outperforming ImageNet pretraining.

These studies show that conventional convolutional networks, joint segmentation–classification approaches, biomarker-based methods, three-dimensional multimodal fusion, and domain-specific foundation models have all been investigated on OCTA-500. The subset, image unit, class definition, data-splitting method, and evaluation metrics must be considered when interpreting the results. In particular, accuracy values from studies using a single projection per patient are not equivalent to image-level results obtained by evaluating hundreds of slices from the same patient.

2.7 Comparison of Literature Studies

Previous studies have reported high performance on retinal images; however, dataset structure, class distribution, splitting method, and evaluation level substantially affect the results. In studies using random image-level splitting, similar slices from the same patient may be distributed across different sets. Overall accuracy may also conceal low recall in minority classes. To reduce these risks, the reported configurations in this study use the

Table 1: Selected studies on disease classification using OCTA-500

Study	Classes	Input	Method	Reported result
Lin et al. [11]	NORMAL, DR, AMD	3 mm and 6 mm OCTA en-face projections	FAZ segmentation and ResNeSt50-based joint learning	Accuracy %94.23; Macro-F1 %91.69
Aydin et al. [12]	NORMAL, DR, AMD; also four classes	OCT and OCTA projections	Comparison of ResNet50 and RETFound	RETFound-OCT highest for three classes; ResNet50-OCTA above ResNet50-OCT
Zhang et al. [13]	Seven disease classes	Three-dimensional OCT and OCTA with projections	Learnable projection and multimodal feature fusion	Accuracy %86.90; F1 %70.81
Shah et al. [14]	NORMAL, DR, AMD, other	Superficial and deep OCTA projections	Cross-modal transfer with VisionFM and RETFound	Accuracy %81.33 and %80.00
This study	NORMAL, DR, AMD	OCT and OCTA volume slices	Four CNN-based configurations; image- and patient-level evaluation	Results are presented in Section 5

same patient-level split, account for class imbalance, and report patient-level results in addition to image-level results.

This study evaluates configurations based on ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 on OCT and OCTA data for the NORMAL, DR, and AMD classes in the OCTA-3mm subset. Its distinguishing features are patient-level data splitting and joint reporting of image- and patient-level results. Because previous studies used different data, class definitions, and validation protocols, their numerical results are not compared directly.

3 Methodology

This study was designed as a quantitative experimental investigation evaluating model configurations based on different deep learning architectures for three-class retinal disease classification in OCT and OCTA images. The experimental process consists of label preparation, patient-level data splitting, image preprocessing and augmentation, transfer-learning-based model training, checkpoint and decision-parameter selection using the validation set, image- and patient-level evaluation on the test set, and selection of the experimental configurations to be reported based on test classification reports.

3.1 Dataset and Classes

Two retinal imaging modalities, OCT and OCTA, were used in the experiments. The dataset consists of OCT and OCTA image folders organized by patient identifier and a `Textlabels.xlsx` file containing patient information. The numbered BMP files in each patient folder are different slice/image samples from that patient; files in the same folder therefore do not represent independent patients. Because the available data files do not contain an acquisition protocol describing the anatomical position or spatial adjacency of slices, file numbers were treated only as within-folder ordering information, and the images were not assumed to be anatomically consecutive. The label file contains patient identifier, sex, imaged eye, age, and disease class fields. The imaged eye is denoted by OD for the right eye and OS for the left eye.

The label file contains 200 records: 160 NORMAL, 29 DR, 6 AMD, and 5 CNV. Four-class experiments were also conducted at the beginning of the study. However, the CNV class contained only five patients, making it difficult to represent the class with sufficient diversity across patient-level training, validation, and test sets. Stable and adequate classification performance for CNV was not obtained in the preliminary four-class experiments. The low patient count and preliminary results were therefore considered together, and the scope was limited to the NORMAL, DR, and AMD classes. Consequently, 195 patients were included in the three-class experiments. The classes are:

- **NORMAL:** Normal cases without a disease label,
- **DR:** Diabetic retinopathy cases,
- **AMD:** Age-related macular degeneration cases.

The three-class patient group consists of 99 males and 96 females, with 98 right-eye and 97 left-eye records. Patient ages range from 7 to 74 years, with a mean age of 33.07 and a median age of 30. These demographic values are provided to describe the dataset and were not included as additional model inputs.

The OCTA dataset contains 304 images for each of 195 patients, for a total of 59,280 images. On the OCT side, image folders are missing for patients 10400 and 10500. Some

Table 2: Patient and image distribution of the three-class dataset

Class	Labeled Patients	OCT Patients	OCT Images	OCTA Patients	OCTA Images
NORMAL	160	158	47,939	160	48,640
DR	29	29	8,788	29	8,816
AMD	6	6	1,824	6	1,824
Total	195	193	58,551	195	59,280

patients with images have 211 or 276 slices, while the others have 304. Therefore, 58,551 images from 193 patients were included in the OCT experiments.

Table 2 shows a marked class imbalance at both patient and image levels. The NORMAL class constitutes most of the dataset with 160 patients, whereas the AMD class contains only 6 patients. Although hundreds of images per patient increase the image count, they do not increase the number of independent patients. Patient counts and patient-level splitting were therefore emphasized over image counts when interpreting the experimental results.

3.1.1 Image File Characteristics

All samples are 8-bit grayscale BMP images with a resolution of 304×640 pixels. Because the pretrained model weights expect three-channel input, grayscale images were converted to RGB in the training pipeline. This operation does not create new color information; it copies the same grayscale information into three input channels.

Figure 1 shows OCT and OCTA examples from three patients labeled NORMAL, DR, and AMD. The 152 .bmp files were selected from both image folders of patient 10301 for NORMAL, 10306 for DR, and 10310 for AMD. Using the same file number only ensures consistent within-folder ordering; because no additional anatomical matching information is available, the OCT and OCTA images are not claimed to correspond to the same anatomical location. The examples illustrate the data format, visual diversity among classes, and appearance differences between the two modalities. No clinical finding or disease-stage interpretation was made from these images.

3.2 Patient-Level Data Splitting

Mixing different images from the same patient across training and test sets may cause the model to learn patient-specific characteristics and artificially inflate test performance. To reduce this risk, the data were split by patient identifiers rather than images. The same `split_3sinif.json` file was used in all reported model configurations to keep evaluation patients fixed.

The OCTA experiments contain 132 patients and 40,128 images in the training set, 24 patients and 7,296 images in the validation set, and 39 patients and 11,856 images in the test set. The OCT experiments use 39,703 images from 131 training patients with available

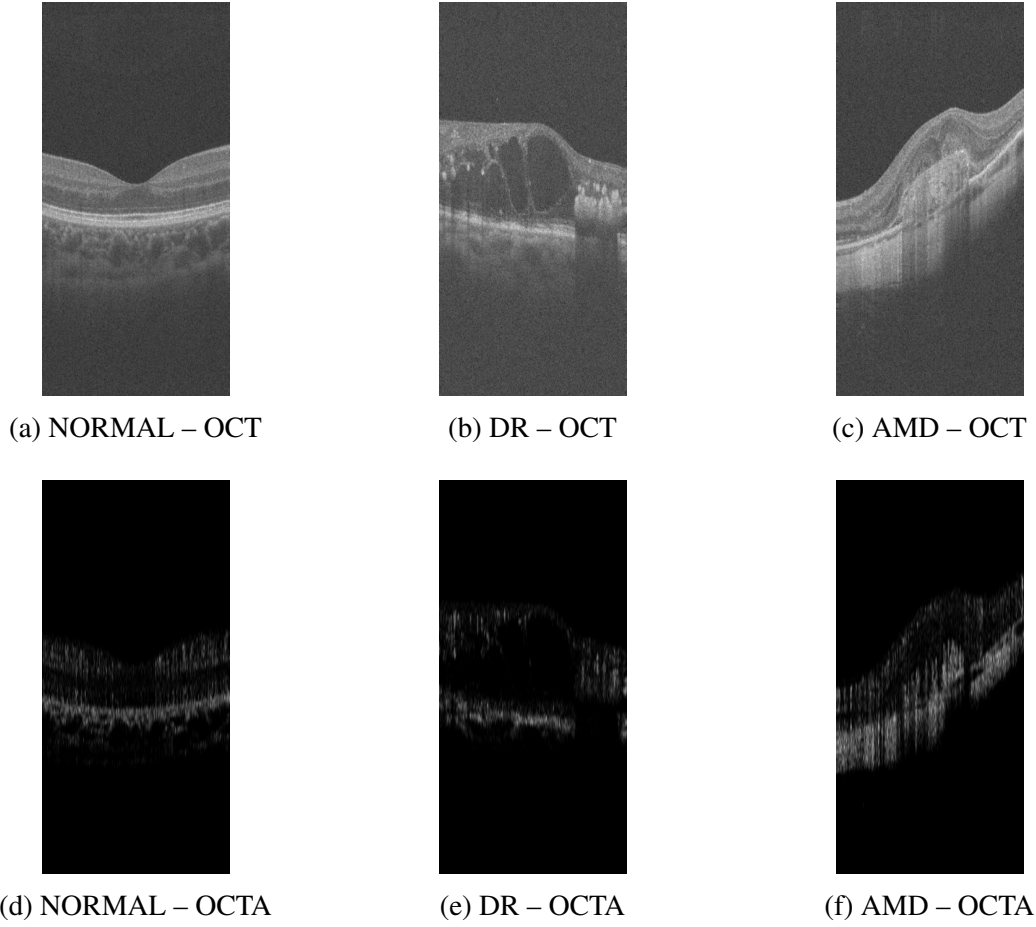


Figure 1: Example OCT and OCTA images from the NORMAL, DR, and AMD classes

images, 7,296 images from 24 validation patients, and 11,552 images from 38 test patients. The absence of overlapping patient identifiers among the training, validation, and test sets was verified programmatically.

3.3 Image Preprocessing and Data Augmentation

Images were resized to 224×224 pixels in the ResNet18, DenseNet121, and EfficientNet-B0 experiments. Images of 300×300 pixels were used with EfficientNet-B3 to benefit from its higher input resolution. Single-channel or differently stored images were converted to RGB format for model input.

Images were normalized using ImageNet means and standard deviations for compatibility with the pretrained ImageNet weights used in transfer learning. Depending on the model and modality, the training set used augmentation operations such as random cropping, horizontal and vertical flipping, limited rotation, brightness and contrast changes, RandAugment, and Random Erasing. Basic validation and test transformations

Table 3: Summary of OCT and OCTA data splits

Modality	Set	Patients	Images
OCTA	Training	132	40,128
OCTA	Validation	24	7,296
OCTA	Test	39	11,856
OCT	Training	131	39,703
OCT	Validation	24	7,296
OCT	Test	38	11,552

consisted of deterministic resizing and normalization. EfficientNet-B3 configurations additionally used deterministic cropping centered on the retinal band. In OCTA ResNet18, probabilities from the original image, horizontal flip, center crop, and fixed rotation were averaged.

3.4 Deep Learning Models

The study uses ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3. ResNet facilitates training deep networks through residual connections [4]. DenseNet uses dense connections that give each layer access to features produced by preceding layers [5]. The EfficientNet family jointly considers network depth, width, and input resolution through compound scaling [6].

Models were initialized with weights pretrained on ImageNet. Their original classification layers were replaced with new layers for the three-class problem. During the initial epochs, the feature-extraction backbone was frozen while the classification head was trained; the backbone layers were subsequently unfrozen and fine-tuned with a low learning rate.

3.5 Management of Class Imbalance

Because patient and image counts differ among classes, a training approach focused only on overall accuracy may favor the majority class. To reduce this effect, weights based on class frequencies were calculated for training samples, and weighted random sampling was used. The experiments also used class-weighted cross-entropy and Focal Loss, which assigns greater importance to difficult samples [15]. MixUp was applied in selected models for regularization [16].

3.6 Training Configuration

The models were developed using PyTorch and Torchvision. All reported configurations used the AdamW optimization algorithm [17], weight decay, learning-rate schedulers, and

early stopping. Automatic mixed precision was used in seven configurations, excluding OCTA ResNet18. ResNet18 and EfficientNet-B3 experiments were configured for a maximum of 35 epochs, while DenseNet121 and EfficientNet-B0 experiments used a maximum of 40 epochs. The mini-batch size was 16 and the random seed was 42 in all experiments.

The distinguishing training and evaluation characteristics of the reported configurations are summarized in Table 4. These differences show that each result belongs to the complete experimental pipeline rather than only to the backbone.

Table 4: Summary of the eight reported model configurations

Type	Backbone	Input	Epochs	Loss	Image Decision	Patient Decision	Distinctive Setting
OCT	ResNet18	224 ²	35	Focal + CE	Threshold	Mean + threshold	Limited OCT-compatible augmentation
OCT	DenseNet121	224 ²	40	Focal + CE	Threshold	Mean + threshold	Limited augmentation
OCT	EfficientNet-B0	224 ²	40	Focal	Argmax	Mean + argmax	RandAugment
OCT	EfficientNet-B3	300 ²	35	Focal + CE	Threshold	Top40 + threshold	Retinal crop, EMA
OCTA	ResNet18	224 ²	35	Focal + CE	Threshold	Mean + threshold	Mean of evaluation transformations
OCTA	DenseNet121	224 ²	40	Focal	Argmax	Mean + argmax	RandAugment
OCTA	EfficientNet-B0	224 ²	40	Focal	Argmax	Mean + argmax	RandAugment
OCTA	EfficientNet-B3	300 ²	35	Focal + CE	Threshold	Mean-Top20 + threshold	Retinal crop, EMA

Because of these differences, the eight results cannot be interpreted as a controlled architectural ablation in which only the backbone changes. A value in the ResNet18 row, for example, is not “the result of the ResNet18 architecture” but the result of a ResNet18-based configuration that includes its corresponding preprocessing, augmentation, loss, optimization, checkpoint, and decision methods. This experimental design cannot isolate how much of the observed difference originates from the backbone and how much from the other components.

3.7 Image- and Patient-Level Decisions

The basic image-level prediction was obtained by selecting the class with the highest probability in the model’s softmax output. Some experiments used class thresholds determined on the validation set to improve minority-class recall.

For patient-level evaluation, image probabilities from the same patient were grouped by patient identifier. Six configurations used the mean probability across all images for each class. In OCT EfficientNet-B3, the NORMAL probability was calculated from all images, while the 40 highest class-specific values were used for DR and AMD. In OCTA EfficientNet-B3, a weight of 0.60 was assigned to the mean probability vector across all images and 0.40 to a vector containing the all-image mean for NORMAL and the means of the 20 highest class-specific probabilities for DR and AMD. Depending on the configuration, argmax or a threshold-based decision determined on the validation set was applied to these patient probabilities.

3.8 Evaluation Metrics

Model performance was evaluated using accuracy, balanced accuracy, precision, recall, F1 score, and Macro-F1. True positive (TP), true negative (TN), false positive (FP), and false negative (FN) values were used to calculate the metrics. In multiclass evaluation, each class was treated as positive in turn and the remaining classes as negative.

3.8.1 Accuracy

Accuracy is the proportion of correctly classified samples:

$$\text{Accuracy} = \frac{\sum_{i=1}^C TP_i}{N}. \quad (1)$$

Here, TP_i denotes the number of correctly classified samples in class i , and N denotes the total number of samples.

Although accuracy provides an intuitive measure of overall performance, it can be misleading in datasets with imbalanced class distributions. In this study, where NORMAL is the majority class, a model may achieve high accuracy despite poor performance on minority classes.

3.8.2 Precision

Precision indicates how many samples predicted as belonging to a class actually belong to that class:

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (2)$$

High precision means that the model produces few false-positive predictions for the corresponding class.

3.8.3 Recall

Recall indicates how many actual samples of a given class are correctly detected:

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3)$$

Low recall is particularly important for disease classes such as DR and AMD because it may indicate that diseased samples are classified as NORMAL.

3.8.4 F1 Score

The F1 score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Because it considers false-positive and false-negative predictions together, this metric complements accuracy in problems with class imbalance.

3.8.5 Macro-F1

Macro-F1 is the arithmetic mean of the F1 scores calculated separately for each class:

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^C F1_i, \quad (5)$$

where C denotes the number of classes. In this study, $C = 3$. Because each class receives equal weight regardless of sample count, Macro-F1 reduces the dominant effect of the NORMAL class and reflects DR and AMD performance in the comparison.

3.8.6 Balanced Accuracy

Balanced accuracy is the mean of class-level recall values:

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{i=1}^C \text{Recall}_i. \quad (6)$$

Because it is less affected by class distribution, balanced accuracy was used as one of the main comparison metrics. In the three-class problem, the recall of each class has equal influence on the result.

3.8.7 Confusion Matrix

A confusion matrix shows the intersections of true and predicted classes. Rows represent true classes and columns represent predicted classes. Diagonal values indicate correct predictions, while off-diagonal values indicate errors between classes. In this study, confusion matrices were used particularly to examine the tendency to classify DR samples as NORMAL or NORMAL samples as diseased.

3.8.8 Image- and Patient-Level Evaluation

All metrics were calculated separately at image and patient levels. At the image level, each OCT or OCTA file was treated as a separate prediction unit. However, images from the same patient are not statistically independent because they may share anatomical and individual characteristics. At the patient level, one prediction was produced using the configuration-specific probability aggregation method described above. Although patient-level evaluation is closer to the clinical decision unit, the number of patients in the test set is much smaller than the number of images. In particular, because the test set contains only one AMD patient, patient-level AMD recall and the associated summary metrics were interpreted cautiously.

Because of the imbalanced class distribution, overall accuracy was not used alone to interpret the results. Balanced accuracy and Macro-F1 were treated as summaries more sensitive to class distribution and were presented together with class-level recall and confusion matrices. These metrics were used to describe the configurations, not as controlled comparison measures proving architectural superiority.

3.9 Implementation Environment and Reproducibility

The experiments were conducted through Python-based Jupyter Notebooks using a local VS Code environment and Google Colab. The data-split file, model weights, training histories, classification reports, confusion matrices, and patient predictions were stored in experimental output directories. Checkpoint selection within each training run was based on validation performance, and threshold values in applicable configurations were also calculated on the validation set. However, among different experiments conducted for the same architecture–modality pair during development, the run selected for detailed reporting was chosen by comparing test classification reports. The test set therefore influenced configuration selection rather than being used only for final evaluation. Consequently, the presented test values are not independent and unbiased estimates of final test performance, but descriptive results of eight configurations selected according to test outcomes. The configurations use the same patient-identifier split; however, because two patients have no OCT folders, the actual patient counts in the OCT and OCTA test sets are not identical.

4 Proposed System and Software Workflow

Rather than developing a single new network architecture, this study implements a modular classification workflow for developing and evaluating OCT and OCTA model configurations based on four convolutional neural network architectures. The system consists of data loading, patient-level splitting, image preprocessing, model training, checkpoint and decision-parameter selection using validation data, test evaluation, and selection of the experimental run to be reported in detail based on test classification reports.

4.1 System Components

The main components of the software workflow are as follows:

1. **Data and label loading:** Patient identifiers and disease labels were read from `Textlabels.xlsx`, and BMP images in the OCT and OCTA folders were matched with the corresponding patient identifiers.
2. **Patient-level splitting:** Training, validation, and test patient identifiers in `split_3sinif.json` were used to prevent images belonging to the same patient from being distributed across different datasets.
3. **Image preparation:** Images were resized to the input dimensions expected by each model, converted to three-channel RGB, and normalized using ImageNet statistics. Random training augmentation was applied to the training set. In OCTA ResNet18 evaluation, probabilities obtained from multiple deterministic transformations were averaged.
4. **Model construction:** ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 were initialized with ImageNet-pretrained weights, and their final classification layers were modified to produce the NORMAL, DR, and AMD classes.
5. **Training and model selection:** Models were optimized on the training set, and the most suitable weights were saved according to validation performance. Weighted sampling, class weights, Focal Loss, and MixUp in selected experiments were applied in the final notebooks to reduce class imbalance.
6. **Testing and reporting:** Each experimental run was evaluated on the test set. Image-level predictions and patient-level predictions obtained by aggregating image probabilities from the same patient were generated. Results were recorded using classification reports and confusion matrices, and the configuration to be reported for each architecture–modality pair was selected by examining test classification reports.

4.2 Software Modules and Outputs

Table 5: Modules of the implemented software workflow

Module	Task	Produced Data/Output
Data preparation	Match labels with patient folders	Image path, patient identifier, and class label
Data splitting	Assign patient identifiers to training, validation, and test sets	Common patient-level split
Training	Optimize model parameters using training data	Model weights and training history
Validation	Select the checkpoint and decision thresholds in applicable experiments	Saved weights and decision parameters
Testing	Produce image- and patient-level predictions	Class probabilities and predicted labels
Evaluation	Calculate performance metrics and error distribution	Classification report and confusion matrix

4.3 Decision-Making Process

At the image level, model softmax probabilities were used in all configurations. An argmax decision was applied in the OCT EfficientNet-B0, OCTA DenseNet121, and OCTA EfficientNet-B0 configurations. The other five configurations used DR and AMD thresholds calculated on the validation set. These thresholds were not recalculated using the test set within the corresponding notebooks. However, the experimental configurations reported in detail were selected by examining test classification reports.

At the patient level, image probabilities belonging to the same patient were grouped by patient identifier. Six configurations used the mean probability across all images for each class. In OCT EfficientNet-B3, the NORMAL probability was calculated from all images, whereas the 40 highest class-specific values were used for DR and AMD. In OCTA EfficientNet-B3, a weight of 0.60 was assigned to the mean probability vector across all images, and 0.40 was assigned to a vector containing the all-image mean for NORMAL and the means of the 20 highest class-specific probabilities for DR and AMD.

In OCTA ResNet18, each image probability was calculated by averaging probabilities obtained from multiple deterministic evaluation transformations. Depending on the configuration, argmax or a threshold-based rule calculated on the validation set was then applied to the resulting patient probabilities to produce one NORMAL, DR, or AMD prediction per patient.

4.4 Unit of Comparison

The unit of comparison in this study is not merely the backbone architecture; it is the complete experimental configuration consisting of the backbone, input size, image transformations, augmentation, sampling, loss function, optimization settings, checkpoint criterion, and decision method. Multiple development experiments with different settings were conducted for the same backbone, and the runs included in detailed reporting were selected by examining test classification reports. The result tables should therefore be read as a technical examination of eight selected implementations, not as a ranking of architectures under equal conditions.

5 Experimental Results and Evaluation

This chapter presents the results of the selected model configuration for each architecture–modality combination by comparing the test classification reports of ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 on OCT and OCTA data. Eight selected configurations consisting of four architectures and two modalities are therefore reported. All configurations were evaluated using the same patient-level split. However, because settings such as input size, augmentation, loss function, decision threshold, and patient-level aggregation method may differ among configurations, the results are not interpreted as the effect of network architecture alone. Since the test set was used in experimental configuration selection, the values in this chapter are presented as descriptive results of the selected configurations on the current dataset rather than as independent final test results. Threshold values in threshold-based experiments were calculated on the validation set.

5.1 OCT Model Results

Each reported OCT model configuration is considered separately. Separate image- and patient-level classification reports are provided for each configuration. The tables show precision, recall, F1 score, and support values from the final notebook outputs. Image-level evaluation is based on predictions produced for each slice, whereas patient-level evaluation is based on a single patient decision formed by aggregating probabilities from slices belonging to the same patient.

5.1.1 OCT ResNet18 Results

In this experiment, ResNet18 initialized with ImageNet weights was trained with 224×224 input for up to 35 epochs. Training transformations included resizing, random cropping, horizontal flipping, brightness–contrast changes, and Random Erasing. Class imbalance was addressed using weighted sampling, class weights, a combination of Focal Loss and cross-entropy with label smoothing, and MixUp. Thresholds determined on the validation set were used for image and patient decisions, and patient probabilities were aggregated by averaging all images.

This ResNet18-based configuration achieved 96.19% balanced accuracy and 95.14% Macro-F1 at the OCT image level. DR recall was 91.12%, and AMD recall was 100%. All 38 patients in the test set were correctly classified at the patient level. However, because the test set contained only six DR patients and one AMD patient, the 100% patient-level results should be interpreted cautiously. The result belongs to the complete training and decision setup described above, not to the ResNet18 backbone alone.

Table 6: OCT ResNet18 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	98.28	97.46	97.87	9,424
DR	87.94	91.12	89.50	1,824
AMD	96.20	100.00	98.06	304
Accuracy	–	–	96.53	11,552
Macro average	94.14	96.19	95.14	11,552
Weighted average	96.59	96.53	96.55	11,552
Balanced accuracy	–	96.19	–	11,552

Table 7: OCT ResNet18 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	100.00	100.00	100.00	31
DR	100.00	100.00	100.00	6
AMD	100.00	100.00	100.00	1
Accuracy	–	–	100.00	38
Macro average	100.00	100.00	100.00	38
Weighted average	100.00	100.00	100.00	38
Balanced accuracy	–	100.00	–	38

5.1.2 OCT DenseNet121 Results

The DenseNet121-based experiment used 224×224 input, up to 40 epochs, a frozen backbone for two epochs, and an AdamW–OneCycleLR setup. Weighted sampling, MixUp, Random Erasing, and a combination of class-weighted Focal Loss and cross-entropy were used in training. RandAugment, retinal-band cropping, and EMA were not applied in this OCT configuration. The reported image- and patient-level decisions are based on thresholds determined on the validation set, and patient probabilities are based on the arithmetic mean across images.

Table 8: OCT DenseNet121 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	96.60	97.54	97.06	9,424
DR	87.81	82.18	84.91	1,824
AMD	92.40	100.00	96.05	304
Accuracy	–	–	95.18	11,552
Macro average	92.27	93.24	92.67	11,552
Weighted average	95.10	95.18	95.12	11,552
Balanced accuracy	–	93.24	–	11,552

Table 9: OCT DenseNet121 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	100.00	96.77	98.36	31
DR	85.71	100.00	92.31	6
AMD	100.00	100.00	100.00	1
Accuracy	–	–	97.37	38
Macro average	95.24	98.92	96.89	38
Weighted average	97.74	97.37	97.45	38
Balanced accuracy	–	98.92	–	38

This DenseNet121-based configuration achieved 95.18% accuracy and 92.67% Macro-F1 at the image level. DR recall was 82.18%, and AMD recall was 100%. At the patient level, DR and AMD recall were 100%, while accuracy was 97.37% because one NORMAL patient was misclassified. Although the argmax patient decision in the same notebook produced 100% accuracy, the table reports the thresholded decision rule determined on the validation set.

5.1.3 OCT EfficientNet-B0 Results

The EfficientNet-B0-based OCT experiment was trained with 224×224 input for up to 40 epochs. RandomResizedCrop, horizontal and vertical flipping, limited rotation, RandAugment, ColorJitter, and Random Erasing were used together in the augmentation pipeline. Weighted sampling, class-weighted Focal Loss, and MixUp were applied, and automatic mixed precision was enabled. The final image decision used argmax, and the patient decision used argmax over the mean probability across all images.

Table 10: OCT EfficientNet-B0 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	95.59	99.14	97.33	9,424
DR	94.66	76.75	84.77	1,824
AMD	99.33	97.70	98.51	304
Accuracy	–	–	95.57	11,552
Macro average	96.53	91.20	93.54	11,552
Weighted average	95.54	95.57	95.38	11,552
Balanced accuracy	–	91.20	–	11,552

Table 11: OCT EfficientNet-B0 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	96.88	100.00	98.41	31
DR	100.00	83.33	90.91	6
AMD	100.00	100.00	100.00	1
Accuracy	–	–	97.37	38
Macro average	98.96	94.44	96.44	38
Weighted average	97.45	97.37	97.27	38
Balanced accuracy	–	94.44	–	38

Although this EfficientNet-B0-based configuration achieved 99.14% image-level recall for NORMAL, DR recall remained at 76.75%. At the patient level, misclassification of one DR patient resulted in 97.37% accuracy and 94.44% balanced accuracy. The results represent this specific training pipeline, which includes RandAugment.

5.1.4 OCT EfficientNet-B3 Results

Unlike the other three OCT configurations, the EfficientNet-B3-based OCT experiment used deterministic cropping centered on the retinal band and 300×300 input. Training continued for up to 35 epochs and used weighted sampling, MixUp, Random Erasing,

a Focal Loss–cross-entropy combination, automatic mixed precision, and exponential moving average (EMA). The image decision was based on thresholds determined on the validation set. At the patient level, the NORMAL probability was calculated from all images, the 40 highest class-specific values were used for DR and AMD, and a threshold-based decision was applied.

Table 12: OCT EfficientNet-B3 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	96.47	98.37	97.41	9,424
DR	92.41	81.41	86.56	1,824
AMD	90.18	99.67	94.69	304
Accuracy	–	–	95.72	11,552
Macro average	93.02	93.15	92.89	11,552
Weighted average	95.66	95.72	95.63	11,552
Balanced accuracy	–	93.15	–	11,552

Table 13: OCT EfficientNet-B3 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	96.77	96.77	96.77	31
DR	83.33	83.33	83.33	6
AMD	100.00	100.00	100.00	1
Accuracy	–	–	94.74	38
Macro average	93.37	93.37	93.37	38
Weighted average	94.74	94.74	94.74	38
Balanced accuracy	–	93.37	–	38

This EfficientNet-B3-based configuration achieved 95.72% accuracy, 93.15% balanced accuracy, and 92.89% Macro-F1 at the OCT image level. DR recall was 81.41%, and AMD recall was 99.67%. At the patient level, one NORMAL and one DR patient were misclassified, resulting in 94.74% accuracy. Differences from the other OCT rows cannot be attributed only to the EfficientNet-B3 backbone because the input size, retinal cropping, EMA, and Top40 aggregation also differ.

5.2 OCTA Model Results

The OCTA configurations are also presented in separate tables. The OCTA test set contains 39 patients, only one of whom belongs to the AMD class; patient-level AMD results should therefore be evaluated in the context of this limited sample.

5.2.1 OCTA ResNet18 Results

The OCTA ResNet18 experiment used 224×224 input and up to 35 epochs. The backbone was frozen for the first two epochs, after which training continued with all layers. Random cropping, horizontal and vertical flipping, rotation, ColorJitter, Random Erasing, weighted sampling, and MixUp were used in the training pipeline. The loss function combined Focal Loss with cross-entropy using label smoothing. During evaluation, probabilities from the original image, a horizontally flipped image, a center crop, and a fixed rotation were averaged. Automatic mixed precision was not used. On the validation set, DR and AMD thresholds were set to 0.40 and 0.50 for image-level decisions and 0.50 and 0.50 for mean patient probabilities, respectively, and were applied unchanged to the test set.

Table 14: OCTA ResNet18 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	93.41	96.54	94.94	9,728
DR	90.26	59.98	72.07	1,824
AMD	51.53	100.00	68.01	304
Accuracy	–	–	91.00	11,856
Macro average	78.40	85.50	78.34	11,856
Weighted average	91.85	91.00	90.73	11,856
Balanced accuracy	–	85.50	–	11,856

Table 15: OCTA ResNet18 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	93.94	96.88	95.38	32
DR	100.00	66.67	80.00	6
AMD	50.00	100.00	66.67	1
Accuracy	–	–	92.31	39
Macro average	81.31	87.85	80.68	39
Weighted average	93.75	92.31	92.28	39
Balanced accuracy	–	87.85	–	39

In this ResNet18-based OCTA configuration, image-level DR recall was 59.98%, lower than for the other classes. At the patient level, four of six DR patients were correctly classified; DR recall was 66.67%, and Macro-F1 was 80.68%. Because image and patient levels use different evaluation units, these values should not be interpreted directly as a measure of improvement. The findings should be considered together with the configuration-specific training and evaluation setup.

5.2.2 OCTA DenseNet121 Results

The DenseNet121-based OCTA experiment used 224×224 input, up to 40 epochs, and a backbone frozen for the first two epochs. Its augmentation pipeline included RandomResizedCrop, horizontal and vertical flipping, rotation, RandAugment, ColorJitter, and Random Erasing. Class imbalance was addressed using weighted sampling, class-weighted Focal Loss, and MixUp. AdamW, OneCycleLR, and automatic mixed precision were applied. The image decision used argmax, and the patient decision used argmax over the mean probability across all images.

Table 16: OCTA DenseNet121 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	95.20	98.55	96.85	9,728
DR	94.96	73.30	82.74	1,824
AMD	80.42	100.00	89.15	304
Accuracy	–	–	94.70	11,856
Macro average	90.19	90.62	89.58	11,856
Weighted average	94.79	94.70	94.48	11,856
Balanced accuracy	–	90.62	–	11,856

Table 17: OCTA DenseNet121 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	96.97	100.00	98.46	32
DR	100.00	83.33	90.91	6
AMD	100.00	100.00	100.00	1
Accuracy	–	–	97.44	39
Macro average	98.99	94.44	96.46	39
Weighted average	97.51	97.44	97.34	39
Balanced accuracy	–	94.44	–	39

This DenseNet121-based OCTA configuration achieved 94.70% accuracy, 90.62% balanced accuracy, and 89.58% Macro-F1 at the image level. At the patient level, only one DR patient was misclassified, resulting in 97.44% accuracy and 96.46% Macro-F1. These are observed values among the reported OCTA configurations and do not demonstrate the superiority of DenseNet121 under controlled conditions.

5.2.3 OCTA EfficientNet-B0 Results

The EfficientNet-B0-based OCTA experiment used 224×224 input and up to 40 epochs. The backbone was frozen for the first two epochs, after which training continued with all

layers. Training transformations consisted of RandomResizedCrop, horizontal and vertical flipping, rotation, RandAugment, ColorJitter, and Random Erasing. Weighted sampling, class-weighted Focal Loss, MixUp, AdamW, OneCycleLR, and automatic mixed precision were applied. Image probabilities and mean patient probabilities were classified using argmax.

Table 18: OCTA EfficientNet-B0 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	94.64	98.03	96.30	9,728
DR	91.63	70.23	79.52	1,824
AMD	79.58	100.00	88.63	304
Accuracy	–	–	93.80	11,856
Macro average	88.62	89.42	88.15	11,856
Weighted average	93.79	93.80	93.52	11,856
Balanced accuracy	–	89.42	–	11,856

Table 19: OCTA EfficientNet-B0 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	94.12	100.00	96.97	32
DR	100.00	66.67	80.00	6
AMD	100.00	100.00	100.00	1
Accuracy	–	–	94.87	39
Macro average	98.04	88.89	92.32	39
Weighted average	95.17	94.87	94.44	39
Balanced accuracy	–	88.89	–	39

This EfficientNet-B0-based configuration achieved 93.80% accuracy and 88.15% Macro-F1 at the image level. At the patient level, all NORMAL and AMD patients were correctly classified, while two incorrect DR predictions resulted in 66.67% DR recall. The results represent the selected run, not the full range of performance that the same backbone might produce with a different training pipeline.

5.2.4 OCTA EfficientNet-B3 Results

The EfficientNet-B3-based OCTA experiment used retinal-band cropping and 300×300 input. Training continued for up to 35 epochs, with the backbone frozen during the first epoch. Weighted sampling, MixUp, Random Erasing, a Focal Loss–cross-entropy combination, automatic mixed precision, and EMA were applied. Image decisions used validation-set thresholds of 0.425 for DR and 0.15 for AMD. To form patient probabilities,

a weight of 0.60 was assigned to the mean probability vector across all images and 0.40 to a vector containing the all-image mean for NORMAL and the means of the 20 highest class-specific probabilities for DR and AMD. Validation-set thresholds of 0.50 for DR and 0.275 for AMD were applied to these probabilities.

Table 20: OCTA EfficientNet-B3 image-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	94.76	97.92	96.31	9,728
DR	97.69	67.11	79.56	1,824
AMD	49.09	88.82	63.23	304
Accuracy	–	–	92.95	11,856
Macro average	80.51	84.61	79.70	11,856
Weighted average	94.04	92.95	92.89	11,856
Balanced accuracy	–	84.61	–	11,856

Table 21: OCTA EfficientNet-B3 patient-level classification report

Class/Summary	Precision (%)	Recall (%)	F1 (%)	Support
NORMAL	100.00	96.88	98.41	32
DR	100.00	83.33	90.91	6
AMD	33.33	100.00	50.00	1
Accuracy	–	–	94.87	39
Macro average	77.78	93.40	79.77	39
Weighted average	98.29	94.87	96.02	39
Balanced accuracy	–	93.40	–	39

Although this EfficientNet-B3-based configuration achieved 93.40% patient-level balanced accuracy, two false-positive predictions for AMD reduced class precision and Macro-F1. Correct classification of the single AMD patient should not be interpreted as generalizable AMD performance. Because of retinal cropping, EMA, 300×300 input, and mean-Top20 aggregation, this row differs from the other OCTA rows in more than the backbone.

5.3 Confusion Matrices

The following figures show the final image- and patient-level confusion matrices for each model. Rows represent true classes, and columns represent predicted classes.

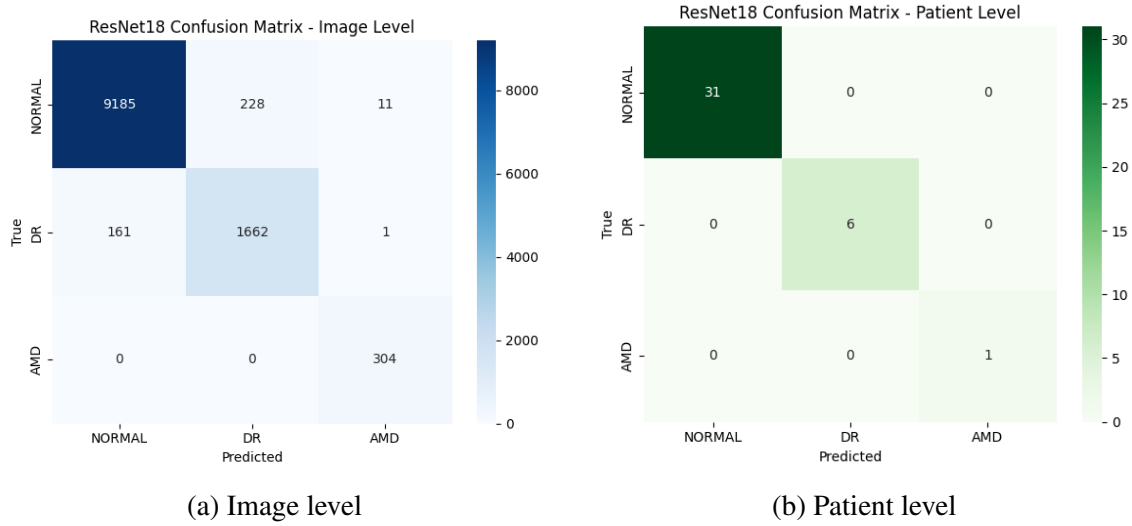


Figure 2: OCT ResNet18 confusion matrices

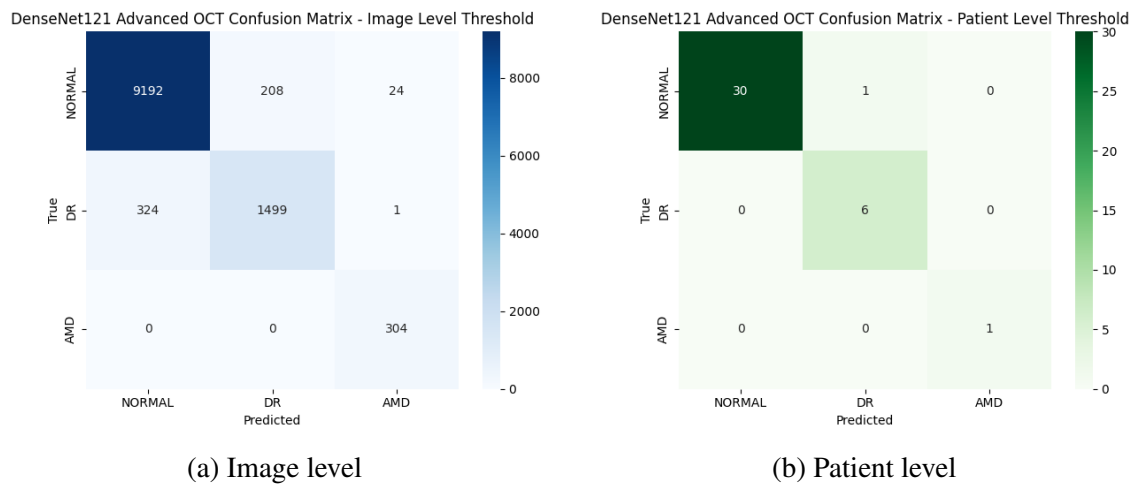
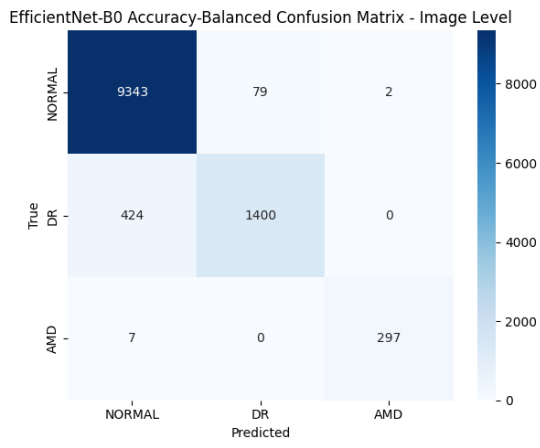
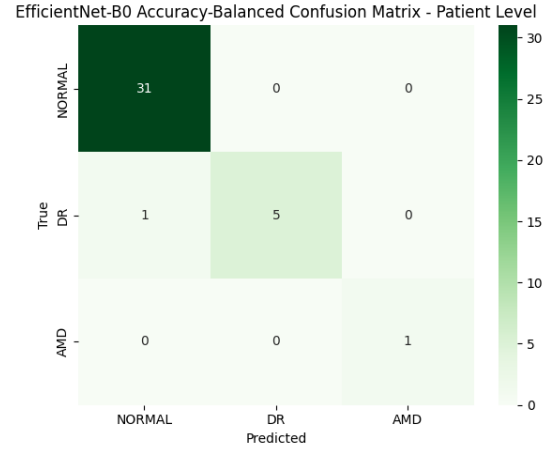


Figure 3: OCT DenseNet121 confusion matrices

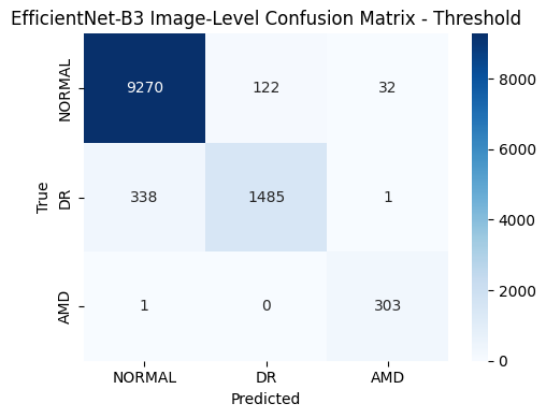


(a) Image level

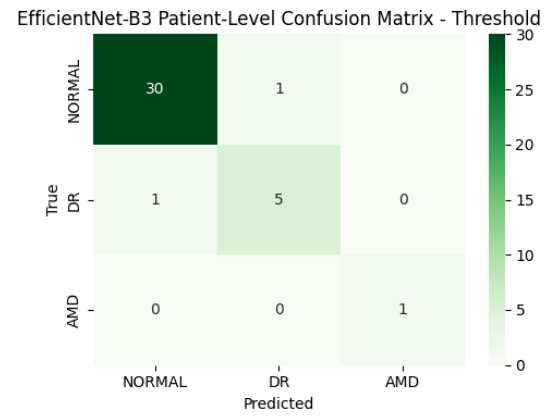


(b) Patient level

Figure 4: OCT EfficientNet-B0 confusion matrices

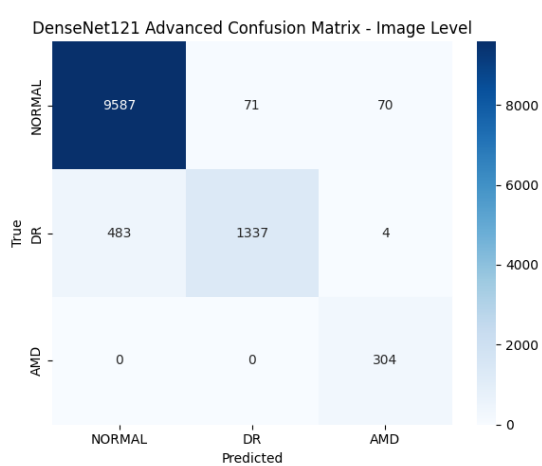


(a) Image level

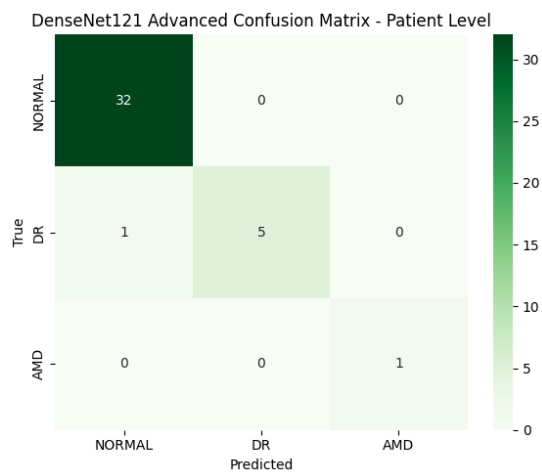


(b) Patient level

Figure 5: OCT EfficientNet-B3 confusion matrices

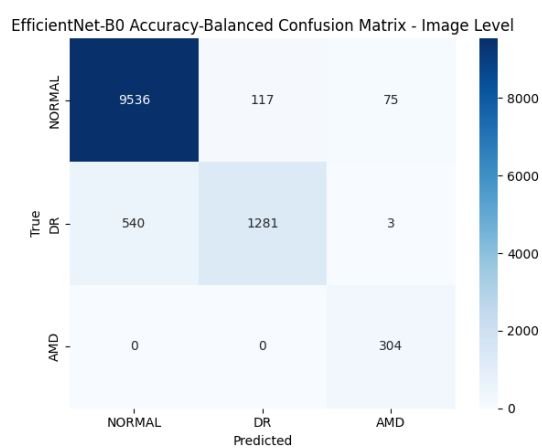


(a) Image level

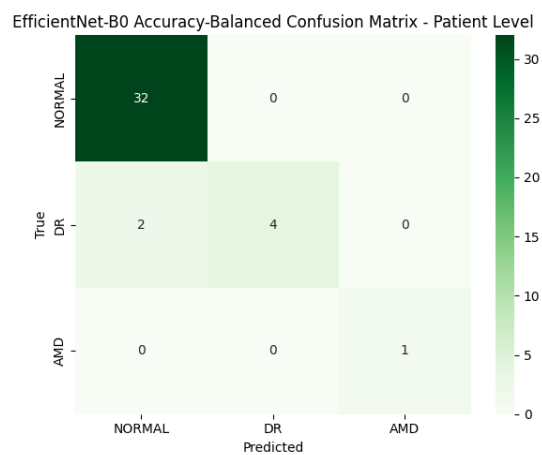


(b) Patient level

Figure 6: OCTA DenseNet121 confusion matrices

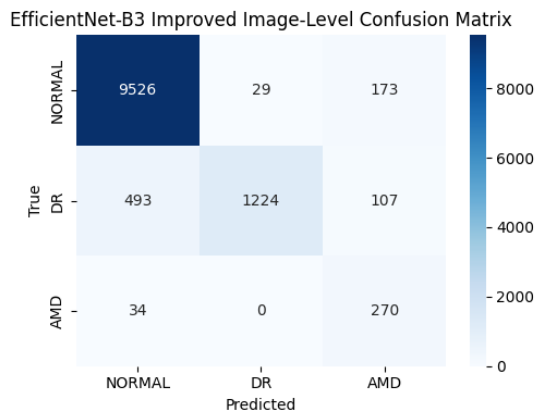


(a) Image level

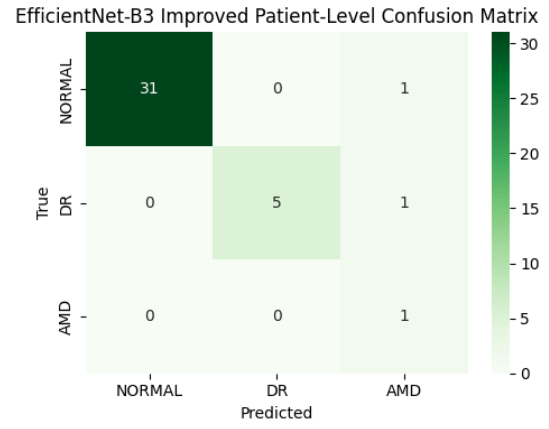


(b) Patient level

Figure 7: OCTA EfficientNet-B0 confusion matrices

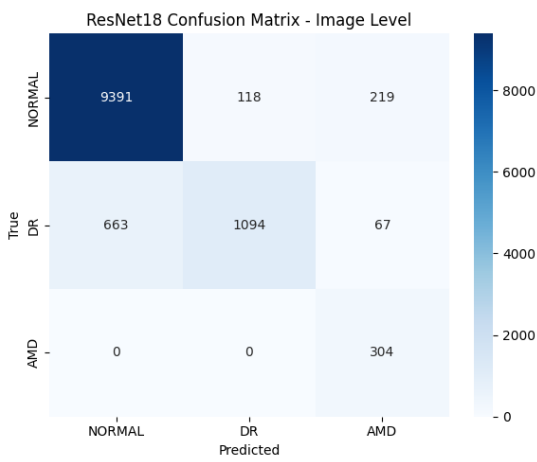


(a) Image level

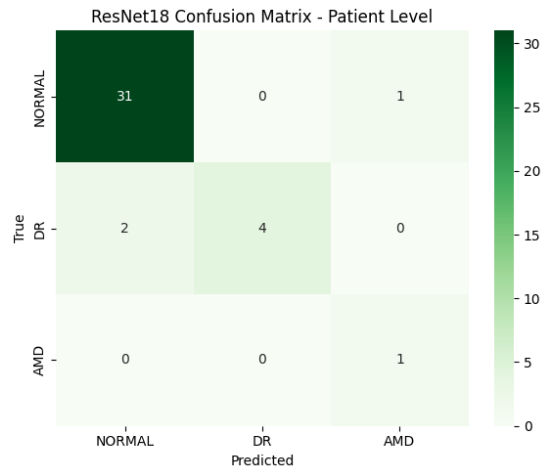


(b) Patient level

Figure 8: OCTA EfficientNet-B3 confusion matrices



(a) Image level



(b) Patient level

Figure 9: OCTA ResNet18 confusion matrices

6 Conclusion and Evaluation

In this graduation project, different model configurations based on ResNet18, DenseNet121, EfficientNet-B0, and EfficientNet-B3 were developed to identify NORMAL, diabetic retinopathy, and age-related macular degeneration classes from OCT and OCTA images. For each architecture–imaging modality combination, the selected configuration was presented by comparing test classification reports, and eight results were examined in detail. Common patient-level training, validation, and test split identifiers were used to prevent patient-level data leakage and to evaluate configurations for each modality on the same available patients.

Among the selected OCT experiments, the ResNet18-based configuration using a thresholded decision rule determined on the validation set achieved 96.53% accuracy, 96.19% balanced accuracy, and 95.14% Macro-F1. Although the argmax test accuracy in the same notebook was 96.61%, the reported class-level table corresponds to the predefined thresholded decision. This finding does not show that the ResNet18 backbone is superior to the other architectures; the run represents a complete configuration that includes OCT-compatible augmentation, a combined loss, weighted sampling, MixUp, and a threshold-based decision rule.

Among the selected OCTA experiments, the DenseNet121-based configuration using RandAugment and an argmax decision achieved 94.70% accuracy, 90.62% balanced accuracy, and 89.58% Macro-F1 at the image level. DR recall in this configuration was 73.30%. In all reported OCTA configurations, the NORMAL class was classified with high recall, whereas DR recall remained lower. The value in the DenseNet121 row is nevertheless the joint result of the backbone and its associated augmentation, loss, sampling, and decision settings.

Aggregating image probabilities at the patient level produced higher summary metrics than image-level evaluation in most experiments. OCT ResNet18 correctly classified all 38 patients in its test set. On the OCTA side, DenseNet121 achieved 97.44% accuracy and 96.46% Macro-F1 across 39 patients. However, because the OCT and OCTA test sets contained only one AMD patient, 100% patient recall for the AMD class should not be considered strong evidence of generalization.

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